

M2S-05: Step 5 — Execute: Migrate Configuration and Agents

Series: M2S — Managed to SaaS Migration | **Notebook:** 5 of 9 | **Phase:** Upgrade | **Step:** Execute | **Created:** March 2026 | **Last Updated:** 07/20/2026

With your SaaS environment prepared, it is time to execute the migration. This step covers deploying configurations via the SaaS Upgrade Assistant in dependency-ordered waves, redirecting OneAgents from Managed to SaaS, and validating data flow after each wave. By the end of this step, all hosts and services will be reporting to your SaaS tenant.

M2S Migration Journey — 3 Phases / 9 Steps

Plan: 1. Discover | 2. Strategize | 3. Design

Upgrade: 4. Prepare | **5. Execute** | 6. Integrate

Run: 7. Expand | 8. Enable | 9. Optimize

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Prerequisites

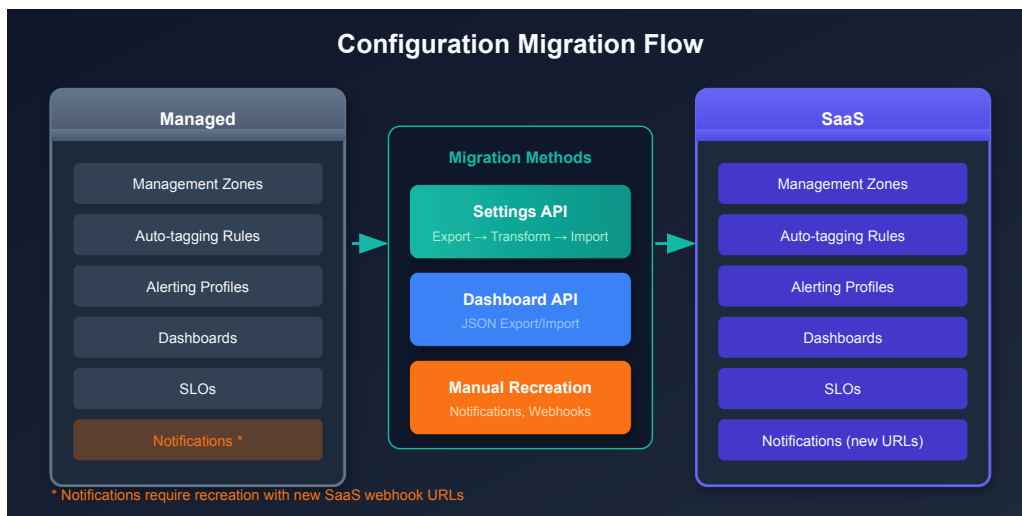
Requirement	Details
Steps 1-4 Complete	Discovery, strategy, design, and preparation all finished
SaaS Tenant	Provisioned with Grail enabled and SaaS Upgrade Assistant installed
ActiveGates Deployed	SaaS-connected ActiveGates operational in all network zones (Step 4)
Network Connectivity	Firewall rules in place for OneAgent-to-SaaS or OneAgent-to-ActiveGate communication
Configuration Archive	Exported from Managed Cluster Management Console
API Tokens	SaaS tokens with <code>settings.write</code> , <code>entities.read</code> , <code>logs.read</code> , <code>metrics.read</code> scopes
Change Windows	Approved change windows for non-production and production waves

Order of Operations — You Are Here

This notebook covers operations **4-6** of the 11-step Order of Operations (see **M2S-02: Step 2 — Strategize**) defined in Step 2:

#	Operation	Status
1	Assess — Inventory current environment	Step 4: Prepare 
2	Provision — SaaS tenant and access (SSO)	Step 4: Prepare 
3	Install — New ActiveGates in parallel	Step 4: Prepare 
4	Migrate — Configuration and integrations	This notebook
5	Rebuild — Dashboards, zones, alerts	This notebook
6	Redirect — OneAgents to SaaS	This notebook
7	Reconnect — Integrations and extensions	Step 6: Integrate

#	Operation	Status
8	Migrate — Remaining config and integrations	Step 6: Integrate
9	Validate — Data flow and performance	Step 9: Optimize
10	Cutover — Full switch to SaaS	Step 9: Optimize
11	Decommission — Managed environment	Step 9: Optimize



1. Configuration Migration Strategy

Configurations must be migrated **before** redirecting OneAgents. When agents start reporting to SaaS, they need to find the correct service detection rules, tagging rules, and monitoring settings already in place. Without this, you lose visibility during the transition.

Migration Order Principle

Configurations FIRST → OneAgents SECOND → Validation AFTER EACH

SaaS Upgrade Assistant Workflow

1. **Export** — Download the configuration archive from Managed Cluster Management Console
2. **Upload** — Import the archive into the SaaS Upgrade Assistant on your target tenant
3. **Review** — The assistant groups configurations by type and flags incompatible items
4. **Edit** — Fix failed or incompatible configurations individually or in bulk
5. **Deploy** — Use Smart Selective Import to push configurations in dependency-ordered waves
6. **Track** — Download the deploy result CSV after each wave

Smart Selective Import

The SaaS Upgrade Assistant supports **Smart Selective Import**, which allows you to deploy configurations incrementally rather than all at once. This is critical for a controlled migration:

Feature	Benefit
Dependency awareness	Automatically identifies prerequisite configurations
Incremental deployment	Deploy one configuration type at a time
Preview before deploy	Review every change before it takes effect
Conflict detection	Flags configurations that already exist in SaaS
Rollback visibility	CSV reports show exactly what was deployed

Important: Always preview changes before deploying. The SaaS Upgrade Assistant shows a diff of what will be created or modified in your SaaS environment.

2. Configuration Wave Deployment

Deploy configurations in four waves, ordered by dependency. Each wave builds on the previous one.

Wave Deployment Order

Wave	Configurations	Rationale
Wave 1	Management zones, auto-tagging rules, host groups	Foundation — everything else depends on these for scoping and targeting
Wave 2	Service detection rules, request attributes, deep monitoring settings	Instrumentation — must be in place before agents report so services are detected correctly
Wave 3	Alerting profiles, anomaly detection, SLOs, maintenance windows	Monitoring — needs entities present to validate thresholds and baselines
Wave 4	Dashboards, calculated metrics, remaining configurations	Visualization — needs data flowing to populate charts and tiles

Wave 1: Foundation

Deploy first because all other configurations reference management zones and tags:

Configuration Type	Why First
Management zones	Scope alerting, dashboards, and access
Auto-tagging rules	Tag entities as they appear — must be ready before agents report
Host groups	Group hosts for configuration targeting

Wave 2: Instrumentation

Deploy before OneAgents migrate so services are detected with the correct naming and grouping:

Configuration Type	Why Second
Service detection rules	Custom service naming and merging must match Managed behavior
Request attributes	Capture rules for request metadata used in filtering and analysis

Configuration Type	Why Second
Deep monitoring settings	Code-level visibility settings (exception rules, method sensors)
Data privacy settings	Masking rules must be active before data flows

Wave 3: Monitoring

Deploy after at least some entities exist in SaaS so the system can validate references:

Configuration Type	Why Third
Alerting profiles	Filter and route alerts to the correct channels
Anomaly detection	Custom thresholds and sensitivity settings
SLOs	Service Level Objectives referencing services and management zones
Maintenance windows	Prevent false alerts during migration activity

Wave 4: Visualization

Deploy last because dashboards and calculated metrics need data to be meaningful:

Configuration Type	Why Last
Classic dashboards	Need entities and data to populate tiles
Calculated metrics	Derive from incoming data — no value without data flow
Remaining configurations	Application detection, RUM settings, etc.

After each wave: Download the deploy result CSV, review failures, fix or document exclusions with justification, then proceed to the next wave.

Entity-Level Configuration Timing

Important: Some entity-level configurations can ONLY be migrated AFTER OneAgents are redirected, because they depend on entities that do not yet exist in the SaaS tenant. Examples: custom settings associated with a specific host or service entity.

Config Type	When to Migrate	Why
Environment-level (management zones, tags, alerting)	BEFORE OneAgent redirect	Independent of entities
Service detection, request attributes	BEFORE OneAgent redirect	Must be in place when services appear
Entity-level custom settings	AFTER OneAgent redirect	Entity must exist in SaaS first
Deep monitoring, container rules	BEFORE OneAgent redirect	Needed for correct instrumentation
Grail bucket routing & retention (OpenPipeline Storage stage)	BEFORE high-volume log-source redirect	Records cannot be moved between buckets after ingest; a bucket's name and class (live/historic) are fixed at creation, while retention and the query-billing model can be changed later

Also note:

- **Process restarts are needed for full-stack mode** — for services to appear in the new SaaS tenant
- **Process restarts are also needed for infra-only mode** — for Application Security vulnerability detection and JMX metrics
- **Minimize delta** with recent configuration changes — this is why the configuration freeze (Step 4) is critical

Log-heavy migration — decide bucket strategy before logs flow. Grail bucket layout, retention, and query-billing model are among the most consequential decisions in a log-driven migration, and most of it is **irreversible after first ingest**: records cannot be moved between buckets, and a bucket's name and class (live/historic) are fixed at creation, while retention and the query-billing model can be changed later. Route logs to the right buckets via the OpenPipeline **Storage** stage on the SaaS tenant *before* redirecting high-volume log sources — otherwise everything lands in `default_logs` and cannot be re-bucketed later. See **ORGNZ-03: Bucket Strategy and Design** and **OPLOGS-04: Buckets & Data Governance** for the full bucket/retention design, and confirm bucket routing is in place as part of Step 4 (Prepare).

3. Manual Configuration Items

The SaaS Upgrade Assistant handles most configurations, but some items require manual recreation due to security constraints or architectural differences.

Items Requiring Manual Recreation

Item	Action	Priority
Credentials Vault	Re-enter all credentials in SaaS Credentials Vault — secrets cannot be exported	Before Wave 3 (integrations need credentials)
API tokens	Generate new tokens with minimal required scopes	Before OneAgent migration
Webhook endpoints	Create new notification integrations with SaaS-appropriate URLs	Before Wave 3 (alerting needs endpoints)
Synthetic private locations	Deploy new Synthetic ActiveGates in SaaS, recreate locations	After ActiveGate deployment
Network zones	Already created in Step 4 (Prepare) — verify they match Managed topology	Verify before OneAgent migration
Custom extensions (1.0)	Rebuild as Extensions 2.0 — EF1 is deprecated	Plan separately (may be post-migration)

Item	Action	Priority
Plugin-based integrations	Migrate to Extensions 2.0 or ActiveGate extensions	Plan separately

Credentials Vault Checklist

For each credential in Managed, recreate in SaaS:

Credential Type	Examples	Notes
Username/password	Database monitoring, SNMP	Must be re-entered manually
Token/key	Third-party integration API keys	Generate new keys if possible
Certificate	mTLS, client certificates	Upload new certificates

API Token Strategy

Create purpose-specific tokens in SaaS with minimal scopes:

Token Purpose	Required Scopes
OneAgent installation	<code>InstallerDownload</code>
Configuration API	<code>settings.read</code> , <code>settings.write</code>
Entity queries	<code>entities.read</code>
Metrics/logs queries	<code>metrics.read</code> , <code>logs.read</code>
AutomationEngine (Workflows)	OAuth client scopes <code>automation:workflows:read</code> , <code>automation:workflows:write</code> (not a classic API-token scope)

Important: Never reuse Managed API tokens. Generate fresh tokens in SaaS with the principle of least privilege.



4. OneAgent Migration

Once configurations are deployed, redirect OneAgents from Managed to SaaS. There are three migration methods — choose based on your requirements.

Migration Methods

Method	Command	Use When
Reconfigure (preferred)	<code>oneagentctl --set-server + --set-tenant-token + restart</code>	Standard migration — preserves host identity, minimal downtime
Parallel install	Install SaaS agent alongside Managed	Zero-gap requirements — both agents report simultaneously during transition
Reinstall	Uninstall Managed agent, install SaaS agent	Clean start needed — resets all agent state

Reconfigure Method (Recommended)

The reconfigure method is preferred because it preserves host identity and requires only a brief restart:

```
# 1. Set new server endpoint
oneagentctl --set-server="https://{tenant-id}.live.dynatrace.com:443/communication"

# 2. Set new tenant token (from SaaS: Settings &gt; Deployment &gt; Installer download)
oneagentctl --set-tenant-token="{tenant-token}"

# 3. Set network zone (if applicable – must match zones created in Step 4)
oneagentctl --set-network-zone=&lt;zone-name&gt;

# 4. Restart OneAgent
# Linux:
systemctl restart oneagent
# Windows:
Restart-Service -Name "Dynatrace OneAgent"
```

If Using ActiveGate Routing

When OneAgents connect through ActiveGates rather than directly to SaaS:

```
# Point to ActiveGate instead of SaaS directly
oneagentctl --set-server="https://{activegate-host}:9999/communication"
oneagentctl --set-tenant-token="{tenant-token}"
oneagentctl --set-network-zone=&lt;zone-name&gt;
systemctl restart oneagent
```

Critical: Restart Application Processes

Warning: Application processes **MUST** be restarted after OneAgent migration. Without a restart, you get host-level metrics but **NO distributed tracing**, **NO service detection**, and **NO code-level visibility**.

What Works Without Restart	What Requires Restart
Host CPU, memory, disk metrics	Distributed tracing (PurePaths)
Network metrics	Service detection and naming
Process-level metrics	Code-level exception analysis
Log monitoring	Request attributes capture

What Works Without Restart	What Requires Restart
	Method-level hotspots

Plan application restarts as part of your migration wave — coordinate with application teams for scheduled restart windows.

Parallel Install Method

For environments requiring zero monitoring gaps:

1. Install the SaaS OneAgent alongside the existing Managed agent
2. Validate data flows to SaaS for the configured period
3. Uninstall the Managed agent once SaaS is confirmed
4. Restart application processes

Note: Running two agents simultaneously doubles the CPU and memory overhead on the host. Use this method only when monitoring continuity is a hard requirement.

Tip — OneAgent Attribute Enrichment (1.333+): During agent redirect, consider adding primary tags and fields at the same time using `oneagentctl --set-host-tag`. This enriches all telemetry at the source with `primary_tags.environment`, `dt.security_context`, `dt.cost.costcenter`, etc. — eliminating the need for some server-side auto-tagging rules. See [docs](#).

Redirecting Containerized and Serverless Workloads

The `oneagentctl` reconfigure method above applies to **host-based OneAgents** (VMs and bare metal). Containerized and serverless workloads have no long-lived host to reconfigure, so the redirect happens through the deployment surface — the operator custom resource, the image build, or the exporter endpoint — not through `oneagentctl`.

Surface	Redirect mechanism	Restart / redeploy
Host VM (GCE / EC2 / on-prem)	<code>oneagentctl --set-server + --set-tenant-token (+ --set-network-zone)</code> , as above	Restart the OneAgent service and restart app processes to re-inject deep monitoring
Kubernetes / GKE (Dynatrace Operator / DynaKube)	<code>spec.apiUrl</code> is immutable (Operator 1.3.0+): delete the DynaKube custom resource and its token secret, recreate both against the new <code>apiUrl</code> , then roll the workloads	Rolling pod restart adopts the new tenant; no node restart
Cloud Run — Java / Node.js (OneAgent)	OneAgent tenant credentials are baked in as build-time image arguments (<code>DT_API_URL</code> , <code>DT_API_TOKEN</code>): rebuild the image against the new tenant and deploy a new revision	The new revision <i>is</i> both the redirect and the re-injection
Cloud Run — other runtimes / OTLP apps	Repoint the OTLP exporter endpoint and API token, then deploy a new revision	New revision

Key difference from VMs: On containerized and serverless surfaces the redirect and the deep-code re-injection are the **same action** — the rolled pods or the new revision. There is no separate "now also restart your applications" step the way there is on a VM, because the workload is recreated from scratch on every deploy.

Cloud Run runtime caveat: OneAgent monitoring of Cloud Run managed is currently limited to **Java and Node.js**; other runtimes (Go, Python, .NET) report via **OTLP** rather than OneAgent — verify current runtime support before planning the redirect. Cloud Run instances appear on the **Hosts** page, and first-generation execution environments do not expose CPU/memory metrics.

5. Migration Wave Execution

Execute the OneAgent migration in waves, starting with non-production and progressing to production.

Wave Execution Order

Wave	Environment	Hosts	Purpose
Wave A	Development/Test	5-10 hosts	Validate process, identify issues
Wave B	Non-production (all)	All non-prod	Confirm at scale, validate services
Wave C	Production (pilot)	10-20% of prod	Validate production behavior
Wave D	Production (full)	Remaining prod	Complete migration

Per-Wave Procedure

For each wave, follow this sequence:

1. **Pre-check** — Verify SaaS ActiveGates are healthy and configurations are deployed
2. **Reconfigure agents** — Run `oneagentctl` commands (manually or via automation)
3. **Restart OneAgent** — Restart the agent service on each host
4. **Verify host check-in** — Confirm hosts appear in SaaS within 5 minutes
5. **Restart applications** — Coordinate with application teams for process restarts
6. **Validate data flow** — Run the DQL queries below to confirm metrics, traces, and logs
7. **Document results** — Record successes, failures, and lessons learned
8. **Apply lessons** — Adjust procedures for subsequent waves

Automation for Bulk Migration

For large environments, use automation tools to execute the reconfigure commands across many hosts:

```
# Ansible example – reconfigure OneAgent on all hosts in a group
ansible-playbook -i inventory/nonprod migrate_oneagent.yml \
  --extra-vars "tenant_url=https://{tenant-id}.live.dynatrace.com:443/communication \
    tenant_token={token} \
    network_zone=datacenter-east"
```

Post-Wave Validation

After each wave, run the following DQL queries against your SaaS tenant to verify data flow.

Verify Hosts Reporting to SaaS

```
// Count hosts reporting to SaaS – compare against expected count from
migration wave
fetch dt.entity.host
| summarize hostCount = count()

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | summarize hostCount = count()
```

Check for Metric Data Gaps

```
// Verify CPU metrics are flowing – hosts with data in the last hour
timeseries avgCpu = avg(dt.host.cpu.usage), from:-1h, by:{dt.entity.host}
| fieldsAdd hasData = arrayAvg(avgCpu)
| filter isNotNull(hasData)
| summarize hostsWithData = count()
```

Verify Services Discovered

```
// Count discovered services – should grow as applications restart
fetch dt.entity.service
| summarize serviceCount = count()

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes SERVICE
// | summarize serviceCount = count()
```

Check Distributed Traces Flowing

```
// Verify spans are arriving – traces require application process restart
fetch spans, from:-1h
| summarize spanCount = count(), by:{time_bucket = bin(timestamp, 5m)}
| sort time_bucket desc
| limit 12
```

Verify Log Ingestion

```
// Confirm log data is flowing from migrated hosts
fetch logs, from:-1h
| summarize logCount = count(), by:{time_bucket = bin(timestamp, 5m)}
| sort time_bucket desc
| limit 12
```

Compare Expected vs Actual Host Count

After each wave, compare the expected host count (from your migration plan) against actual hosts reporting:

```
// Hosts grouped by OS type – verify all expected host types are present
fetch dt.entity.host
| summarize hostCount = count(), by:{osType}
| sort hostCount desc

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | summarize hostCount = count(), by:{osType}
// | sort hostCount desc
```

Tip: If span counts are zero but host metrics are flowing, application processes have not been restarted yet. Coordinate restarts before declaring the wave complete.

6. SaaS Upgrade Assistant Deploy Results

After each configuration wave deployment, download and review the deploy result CSV from the SaaS Upgrade Assistant.

Result Categories

Status	Meaning	Action
Deployed	Configuration successfully created or updated in SaaS	No action needed
Skipped	Configuration already exists in SaaS (no overwrite)	Verify existing config is correct
Failed	Configuration could not be deployed (incompatible or error)	Fix and retry, or document exclusion
Excluded	Manually excluded from deployment	Document justification

Handling Failed Configurations

For each failed configuration:

1. **Identify the failure reason** — The CSV includes error details
2. **Determine if fixable** — Can the configuration be edited in the SaaS Upgrade Assistant?
3. **Fix and retry** — Edit the configuration and redeploy
4. **Or exclude with justification** — If the configuration is no longer needed or will be rebuilt differently in SaaS

Common Failure Reasons

Failure	Cause	Resolution
Entity reference not found	Entity ID changed between Managed and SaaS	Update entity reference or let it auto-resolve after agents report
Unsupported configuration type	Legacy feature not available in SaaS	Rebuild using SaaS-native equivalent

Failure	Cause	Resolution
Duplicate name conflict	Configuration with same name already exists	Rename or merge with existing
Permission denied	Missing IAM policy	Add <code>upgrade-assistant:environments:write</code> policy

Archiving Deploy Results

Archive all deploy result CSVs for compliance and audit purposes:

```
deploy-results/
  wave-1-foundation-YYYY-MM-DD.csv
  wave-2-instrumentation-YYYY-MM-DD.csv
  wave-3-monitoring-YYYY-MM-DD.csv
  wave-4-visualization-YYYY-MM-DD.csv
```

Best Practice: Store deploy result CSVs in your migration project repository alongside your migration runbook. This creates an auditable record of every configuration change.

7. Step Completion Checklist

Before proceeding to Step 6, confirm that you have completed each item:

Checkpoint	Status
Wave 1 deployed: management zones, auto-tagging rules, host groups	[]
Wave 2 deployed: service detection, request attributes, deep monitoring	[]
Wave 3 deployed: alerting profiles, anomaly detection, SLOs	[]
Wave 4 deployed: dashboards, calculated metrics, remaining configs	[]
Manual items recreated (credentials vault, API tokens, webhook endpoints)	[]
All deploy result CSVs downloaded and archived	[]

Checkpoint	Status
Non-production OneAgents migrated and validated	[]
Production OneAgents migrated and validated	[]
Application processes restarted (all waves)	[]
Data flow confirmed: host metrics flowing	[]
Data flow confirmed: distributed traces flowing	[]
Data flow confirmed: log ingestion active	[]

Next Step

M2S-06: Step 6 — Integrate — Reconnect all integrations and extensions. Migrate notification channels, rebuild Extensions 2.0, configure cloud integrations, and validate end-to-end alerting workflows.

Additional Resources

- [SaaS Upgrade Assistant Documentation](#)
- [OneAgent Configuration via Command Line](#)
- [Migrate Dynatrace Operator to a New Environment](#)
- [Monitor Google Cloud Run \(Managed\)](#)
- [Network Zones Configuration](#)
- [Extensions 2.0 Framework](#)

Summary

In Step 5, you:

- Deployed configurations to SaaS in four dependency-ordered waves using the SaaS Upgrade Assistant

- Recreated manual configuration items (credentials vault, API tokens, webhook endpoints)
- Migrated OneAgents from Managed to SaaS using the reconfigure method
- Restarted application processes for full distributed tracing and service detection
- Validated data flow (metrics, traces, logs) after each migration wave
- Archived all deploy result CSVs for compliance

Key Takeaway: Migrate configurations before agents, validate after every wave, and always restart application processes. Without application restarts, you see host metrics but lose distributed tracing and service detection.

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